



Heart Disease Risk Analysis

Exploratory Data Analysis • UCI Cleveland Heart Disease Dataset

1,025

Patients

14

Variables

51%

Disease Rate

0

Missing Values

Tools: Python • Pandas • Matplotlib • Seaborn • SciPy

DATASET OVERVIEW



1,025

Total Patients

4 clinical databases:
Cleveland, Hungary,
Switzerland, Long Beach



14

Variables

Age, sex, cholesterol,
heart rate, chest pain
type & more



0

Missing Values

Fully clean dataset,
ready for analysis
without imputation



51%

Disease Rate

525 with disease
500 without disease
(balanced dataset)

Key Variables

Variable	Description
age	Patient age in years
sex	Gender (1=Male, 0=Female)
cp	Chest pain type (0–3)
thalach	Maximum heart rate achieved
chol	Serum cholesterol (mg/dL)
target	Heart disease: 1=Yes, 0=No



Dataset Limitations

- Data collected in 1988 — may not reflect modern patients
- Primarily Western demographics
- Southeast Asian populations not represented
- Findings may not generalise globally

Source: UCI Machine Learning Repository — Cleveland Heart Disease Dataset (via Kaggle)

KEY STATISTICS

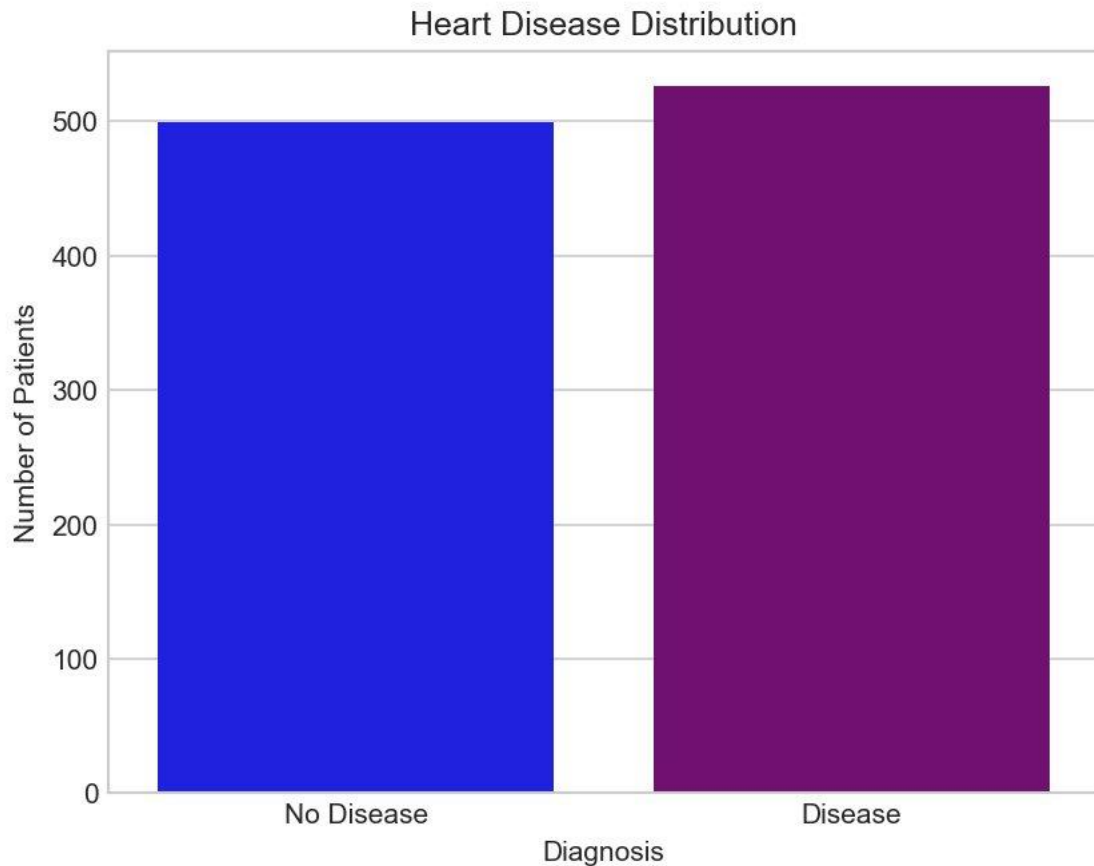
Comparing patients with and without heart disease across key variables

Metric	No Disease	Disease	Insight
Average Age	56 years	52 years	Small gap — age alone is a weak predictor
Avg Cholesterol	251 mg/dL	241 mg/dL	Higher in healthy patients ⚠️ Counterintuitive
Avg Max Heart Rate	143 bpm	158 bpm	Higher heart rate = stronger disease signal
Exercise Angina	20%	44%	More than double — strong disease indicator
Typical Chest Pain	More common	Less common	Opposite of expectation ⚠️ Misdiagnosis risk

🔑 **Top Predictors of Heart Disease:**

cp (chest pain type) • thalach (max heart rate) • exang (exercise angina) • oldpeak (ST depression)

FINDING 1 — DISEASE PREVALENCE



Balanced Dataset

500 healthy vs 525 disease patients — a near-perfect 49/51 split, ideal for unbiased analysis.



Why It Matters

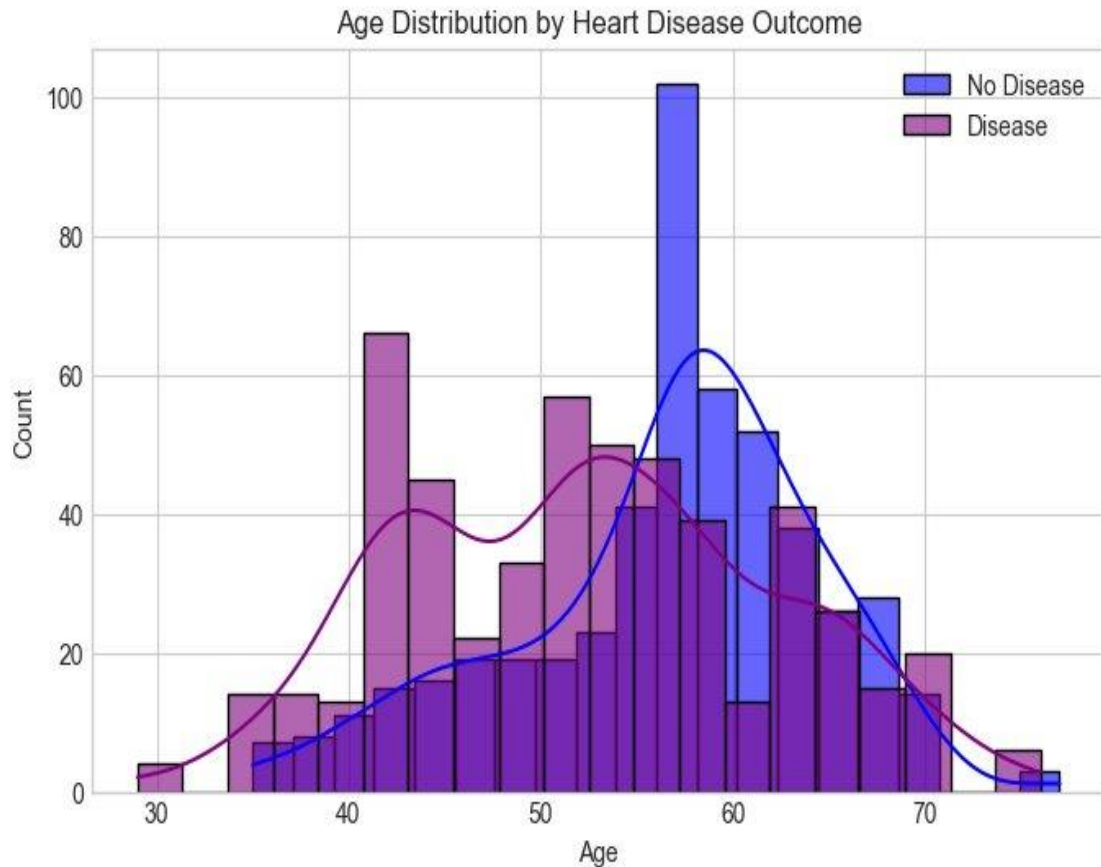
Balanced data means findings aren't skewed toward one group, making statistical results more reliable.



Takeaway

Heart disease affects just over half of patients in this clinical sample — a significant public health concern.

FINDING 2 — AGE DISTRIBUTION BY OUTCOME



Disease Peaks at Age 58

While average age with disease is 52, the visual distribution peaks around 58 — middle-to-late age carries higher risk.

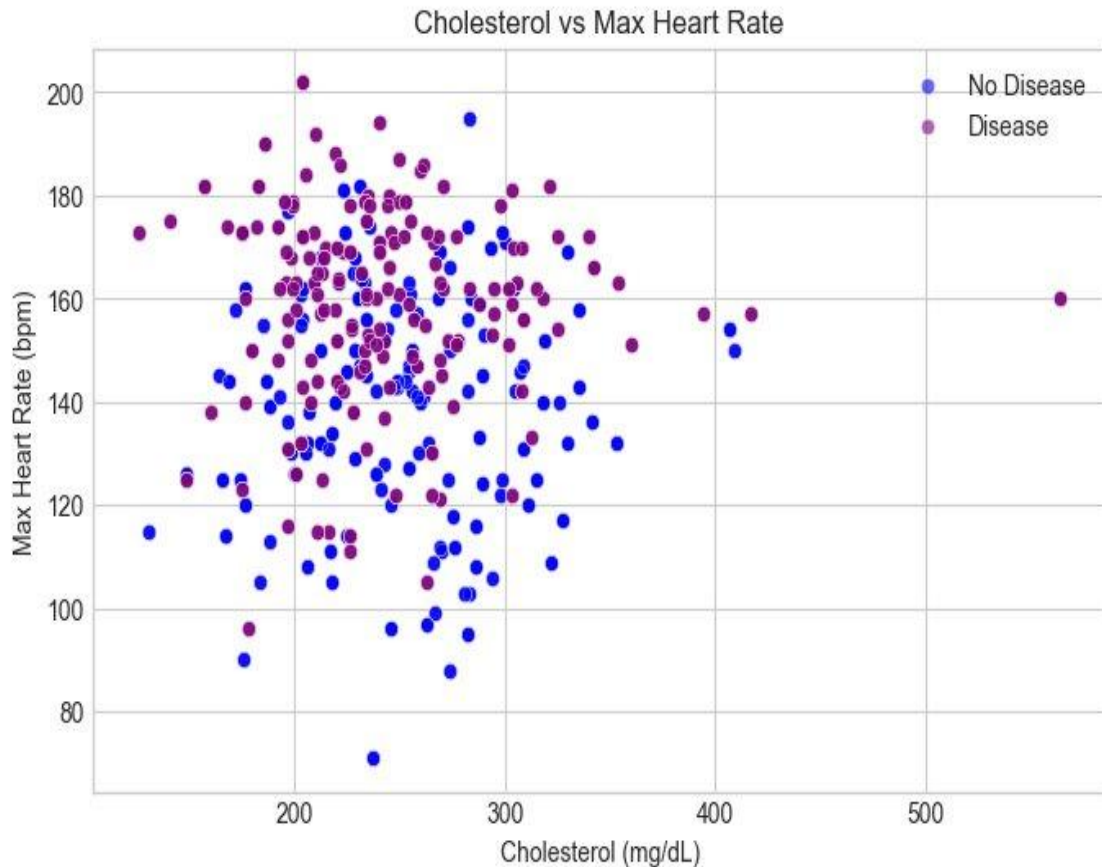
Mean Can Mislead

Statistics showed only a 4-year gap (52 vs 56). The histogram reveals the full picture that a single number hides.

Key Insight

Age contributes to risk but is not the dominant predictor. Heart rate and chest pain type are stronger signals.

FINDING 3 — CHOLESTEROL vs MAX HEART RATE



🧐 Cholesterol Counterintuitive

Disease patients (purple) cluster at lower cholesterol — contradicting the assumption that high cholesterol always means disease.

❤️ Higher Heart Rate = More Risk

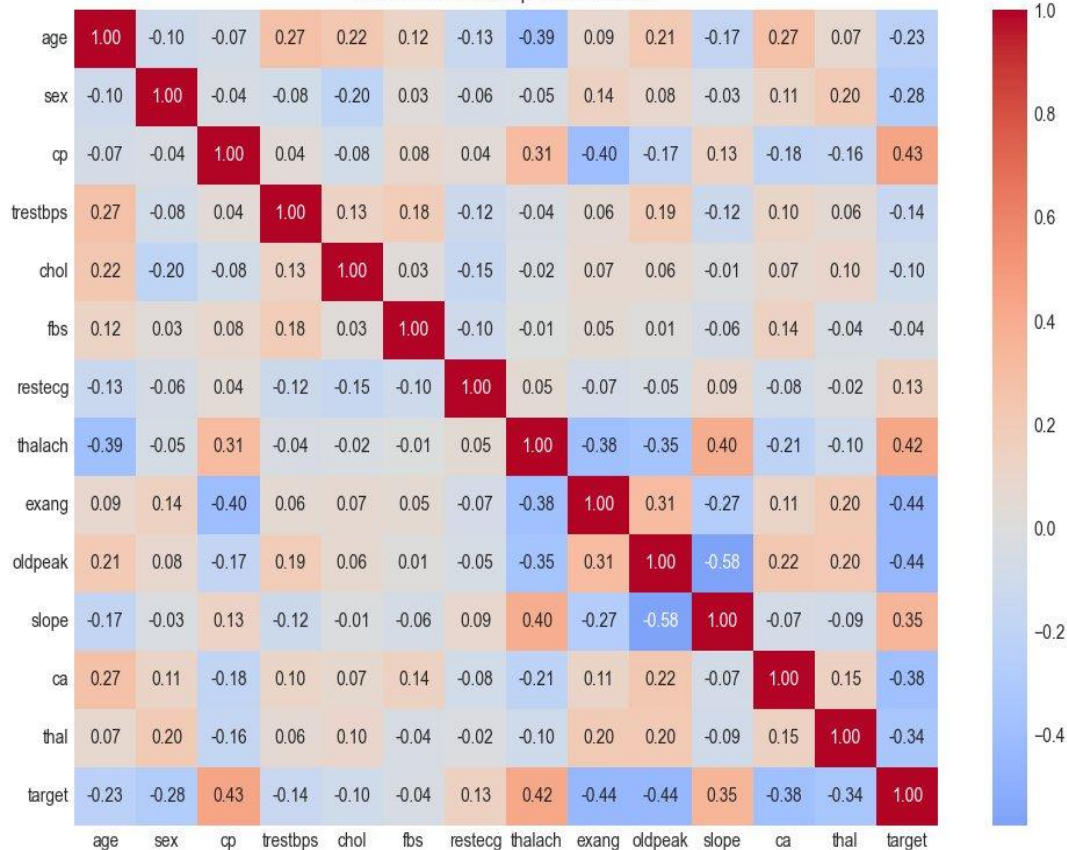
Disease patients cluster at higher max heart rates (160–200 bpm), confirming thalach as a top predictor.

🔬 Multivariate Thinking

No single variable tells the whole story. Heart disease is driven by a combination of risk factors — not cholesterol alone.

FINDING 4 — FEATURE CORRELATION HEATMAP

Correlation Heatmap of All Features



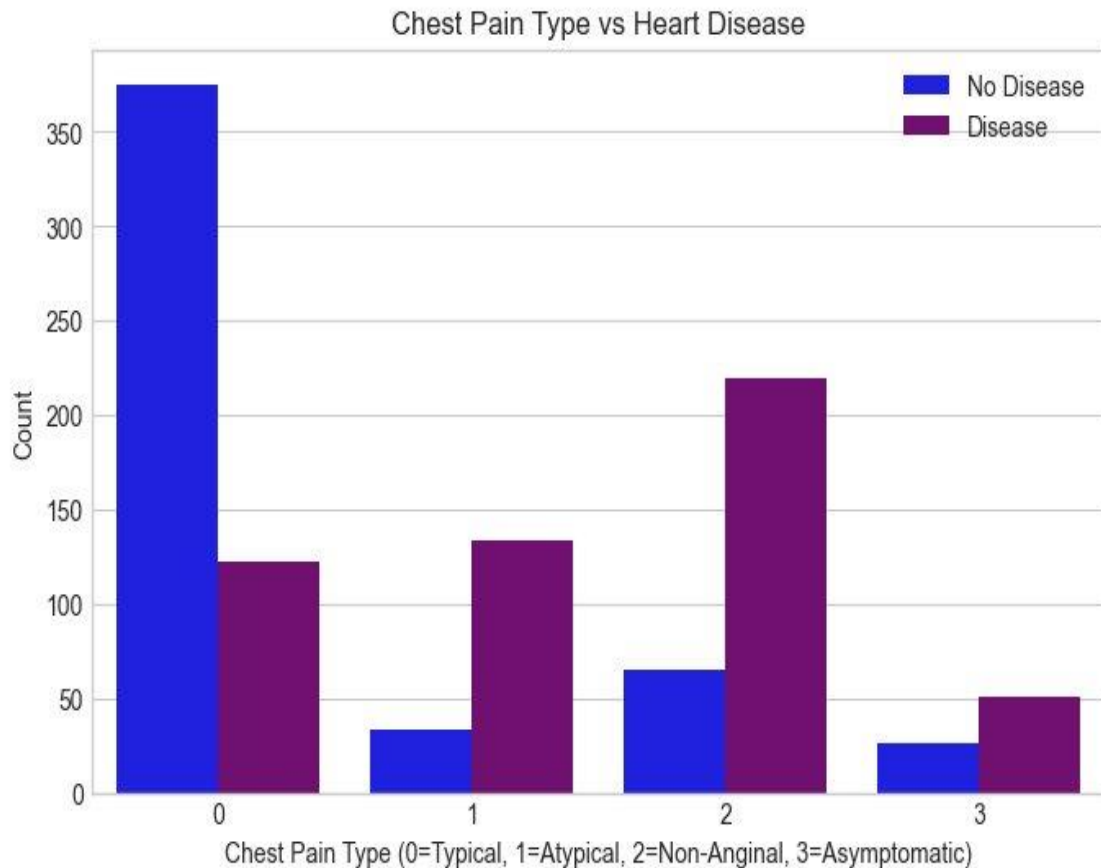
Correlations with Target

Variable	Value	Direction
cp (chest pain)	+0.43	↑ Positive
thalach (max HR)	+0.42	↑ Positive
slope	+0.35	↑ Positive
exang (ex. angina)	-0.44	↓ Negative
oldpeak (ST dep.)	-0.44	↓ Negative
ca (vessels)	-0.38	↓ Negative

How to read correlations:

- +1.0** Variables rise together
- 0.0** No relationship
- 1.0** One rises, other falls

FINDING 5 — CHEST PAIN TYPE vs HEART DISEASE



🧐 Typical Angina Misleads

Patients with typical chest pain (cp=0) were more often healthy — the opposite of what clinical intuition suggests.

⚠️ Non-Anginal is Highest Risk

Non-anginal chest pain (cp=2) had the highest disease count — the most diagnostically important type.

🏥 Clinical Implication

Chest pain type alone cannot diagnose heart disease — a key reason why it is frequently misdiagnosed.

CONCLUSIONS & NEXT STEPS

Key Findings

- 01 Chest pain type and max heart rate are the strongest predictors of heart disease in this dataset.
- 02 Cholesterol is counterintuitively higher in healthy patients — single variables are unreliable predictors.
- 03 Typical angina presents more in healthy patients, highlighting misdiagnosis risk in clinical practice.
- 04 Age contributes to risk but is not dominant — heart disease peaks visually around age 58.



Next Steps



Machine Learning Model

Build a Logistic Regression classifier to predict heart disease using the top correlated features.



Model Evaluation

Evaluate with accuracy, precision, recall and confusion matrix to measure real-world reliability.



Streamlit Web App

Deploy an interactive app so non-technical users can explore the data and predictions visually.



Code on GitHub

Full analysis, charts and methodology:
github.com/hazelkimhyejin/heart-disease-analysis



Thank You

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GitHub: github.com/hazelkimhyejin/heart-disease-analysis



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